

Cutaneous Adverse Reactions Associated with Novel Antihypertensives: A Clinical Case Series of Three Distinct Patients

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ABSTRACT -- Background: Emerging antihypertensive modulators (Cardioril and Corzapan) have demonstrated potent hemodynamic efficacy, yet post-marketing surveillance reveals heterogeneous cutaneous toxicities. We report three distinct individual patient cases evaluated across two UK tertiary centers.

Case Series Summary: **Case 1 (Patient A.J.):** A 45-year-old male (81 kg) treated with Cardioril 20 mg once daily for stage II hypertension developed Erythema Multiforme Major with target lesions, central blistering, and severe oral mucosal erosions on day 22 of therapy, requiring inpatient hospitalization and systemic corticosteroids (serious: true). **Case 2 (Patient B.L.):** A 62-year-old female (64 kg) receiving Corzapan 10 mg daily for essential hypertension presented after 8 weeks with Drug-Induced Subacute Cutaneous Lupus Erythematosus (SCLE) characterized by annular polycyclic scaly plaques, high-titer anti-Ro/SSA (>240 U/mL), and ANA 1:320, resolving fully with drug withdrawal and topical clobetasol (serious: false). **Case 3 (Patient C.M.):** A 38-year-old female (59 kg) administered a single dose of Cardioril 10 mg for mild hypertension experienced life-threatening acute generalized urticaria, severe pruritus, dysphonia, and marked bilateral periorbital angioedema within 90 minutes, successfully rescued with emergency intramuscular epinephrine 0.3 mg and intravenous diphenhydramine 50 mg (serious: true; life-threatening: true).

Conclusion & PV Mandate: Each patient represents an independent ICSR with distinct pharmacodynamic mechanisms requiring individual regulatory reporting.

Keywords: Cardioril; Corzapan; Erythema Multiforme Major; Subacute Cutaneous Lupus Erythematosus; Angioedema; Epinephrine; Pharmacovigilance; Case Splitting.

1. INTRODUCTION & CASE IDENTIFICATION

Adverse drug reactions (ADRs) targeting the skin and mucous membranes vary widely in severity, from self-limited lichenoid eruptions to life-threatening angioedema and bullous dermatoses [1]. Novel antihypertensive agents targeting vascular receptors have entered broad clinical use. Over a 14-month period, our collaborative regional network identified three separate adult patients who experienced distinct, biopsy- or biomarker-proven adverse reactions linked to Cardioril and Corzapan. To assist pharmacovigilance screening and individual case safety report (ICSR) processing, each patient is described independently below.

2. CASE 1: PATIENT A.J. (ERYTHEMA MULTIFORME MAJOR)

Patient Details: Patient A.J. is a 45-year-old male (body weight 81 kg) with newly diagnosed stage II essential hypertension without prior medication allergies. **Exposure:** Cardioril monotherapy was initiated at 20 mg once daily (PO QD). **Clinical Course & Reaction:** On Day 22 of therapy, he presented with a painful, widespread eruption consisting of classic target lesions with central blistering predominating on the palms, forearms, neck, and trunk. In addition, severe oral mucosal erosions with mucosal hemorrhage developed, preventing oral intake. Skin biopsy demonstrated necrotic basal keratinocytes with subepidermal clefting and satellite cell necrosis, diagnostic of **Erythema Multiforme Major**. **Management & Outcome:** Cardioril was permanently discontinued (positive dechallenge). The patient required inpatient hospitalization for 5 days with IV fluids, analgesia, and a short course of oral prednisolone (30 mg/day). Lesions re-epithelialized completely over 18 days without mucosal strictures. **Regulatory Classification:** Serious (Inpatient Hospitalization required).

3. CASE 2: PATIENT B.L. (SUBACUTE CUTANEOUS LUPUS)

Patient Details: Patient B.L. is a 62-year-old female (body weight 64 kg) with a 6-year history of uncomplicated essential hypertension. **Exposure:** Corzapan 10 mg PO daily was instituted following an inadequate response to amlodipine. **Clinical Course & Reaction:** After 8 weeks of continuous therapy, she developed widespread, photosensitive annular polycyclic scaly plaques distributed over the upper chest, back, and extensor arms. Systemic symptoms (fever, arthralgia, pleurisy) were absent. Punch biopsy revealed hyperkeratosis, basal vacuolar degeneration, and band-like dermal lymphocytic infiltration. Serological evaluation showed strongly positive anti-Ro/SSA antibodies (>240 U/mL, ref <7 U/mL) and elevated antinuclear antibodies (ANA 1:320, speckled), confirming **Drug-Induced Subacute Cutaneous Lupus Erythematosus (SCLE)**. Anti-La/SSB and anti-dsDNA were negative. **Management & Outcome:** Corzapan was withdrawn (positive dechallenge) and topical clobetasol 0.05% ointment applied. Lesions resolved completely within 6 weeks, with anti-Ro/SSA titers decreasing to 38 U/mL at 12-week follow-up. **Regulatory Classification:** Non-serious (outpatient management, no hospitalization).

4. CASE 3: PATIENT C.M. (ACUTE ANGIOEDEMA & URTICARIA)

Patient Details: Patient C.M. is a 38-year-old female (body weight 59 kg) with newly detected mild hypertension (144/92 mmHg) and no history of asthma or atopy. **Exposure:** Administered her initial introductory dose of Cardioril 10 mg PO at 09:00. **Clinical Course & Reaction:** Approximately 90 minutes following the initial dose, she developed acute generalized urticaria with intense severe pruritus, rapidly followed by marked bilateral periorbital angioedema, facial swelling, globus sensation, and progressive dysphonia. Emergency medical services were dispatched. On arrival (vital signs: BP 90/58 mmHg, HR 128 bpm, SpO2 93%), she was in respiratory distress with inspiratory stridor. **Emergency Treatment & Outcome:** Rescued with emergency intramuscular epinephrine 0.3 mg (anterolateral thigh), followed by intravenous diphenhydramine 50 mg and methylprednisolone 60 mg. Laryngeal edema and angioedema rapidly resolved within 45 minutes. She was monitored in the emergency resuscitation bay for 8 hours and discharged without inpatient admission. Cardioril was permanently blacklisted. **Regulatory Classification:** Serious (Life-Threatening emergency).

Table 1. Cross-Case Comparative Matrix (3 Distinct Individual ICSR Cases)

Case	Pt / Demog	Drug / Dose	Latency	Reaction / Diagnosis	Seriousness
Case 1	A.J. 45M, 81kg	Cardioril 20 mg QD	Day 22	Erythema Multiforme Major (Target lesions, oral erosions)	Serious (Hospitalized)
Case 2	B.L. 62F, 64kg	Corzapan 10 mg QD	8 weeks	Subacute Cutaneous Lupus (Annular plaques, anti-Ro >240)	Non-serious (Outpatient)
Case 3	C.M. 38F, 59kg	Cardioril 10 mg QD	90 min	Acute Angioedema & Urticaria (Dysphonia, periorbital edema)	Serious (Life-threat.)

5. PHARMACOVIGILANCE & CASE SPLITTING IMPLICATIONS

Under ICH E2D and EMA GVP Module VI, multi-patient case series describing discrete individuals must be processed by automated intake engines as **three independent ICSR safety records** rather than a single aggregated report. Each case possesses unique patient demographics, exposure histories, clinical phenotypes, and regulatory seriousness thresholds (Cases 1 and 3 serious; Case 2 non-serious). Combining these cases would obscure safety signals and compromise signal detection registries.

REFERENCES

1. Roujeau JC. Clinical heterogeneity of severe cutaneous adverse reactions. *Toxicology*. 2005;209(2):123-129.
2. Sontheimer RD. Subacute cutaneous lupus erythematosus: 25-year evolution. *J Invest Dermatol*. 2004;122(2):11-15.
3. Simons FE. Anaphylaxis: Recent advances in assessment and treatment. *J Allergy Clin Immunol*. 2009;124(4):625-636.
4. European Medicines Agency. Guideline on good pharmacovigilance practices (GVP) Module VI. 2017.